Gude Manjunadh

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PROFESSIONAL SUMMARY

Computer Science graduate focused on AI/ML and GenAI engineering with hands-on experience building reliable LLM systems, fault-tolerant agent workflows, evidence-grounded document intelligence, temporal memory retrieval, and reproducible tool-calling evaluations using Python.

WORK EXPERIENCE

Artificial Intelligence Intern - SmartED Innovations Jul 2025 - Sep 2025

USNG LearnSpace EdTech Pvt Ltd (Skill India & NSDC accredited) · 2-month programme · batch "SmartED Jul 15, 2025"

- Month 1 - mentor-led training through live sessions and recorded content covering current industry trends, technical best practices, and professional skills.
- Month 2 - hands-on research and development on minor and major projects against real-world problem statements, in teams, with continuous industry-mentor guidance.
- Completed both components; issued Certificate of Internship Completion (Student ID AI1332, 23 Sep 2025) and Certificate of Course Completion (ID CCC1721, 29 Jan 2026).

MERN Full-Stack Developer Intern - Ethnus Jun 2024 - Jul 2024

- Developed a full-stack online quiz application using MongoDB, Express.js, React.js, and Node.js, with role-based features for students and administrators.
- Implemented REST APIs, authentication, quiz management, dynamic questions, timed sessions, and automatic scoring.
- Designed a responsive interface with streamlined quiz navigation and results, while optimizing MongoDB data retrieval and session management.

Technologies / Skills Used: *MongoDB, Express.js, React.js, Node.js, REST APIs, authentication*

PROJECTS

AgentForge | Fault-Tolerant AI Agent Runtime GitHub

Stack: *Python, LangGraph, FastAPI, PostgreSQL, SQLite, Pydantic, Pytest*

- Reduced duplicate handler executions by **80% (500 to 100)** across 100 simulated actions with five delivery attempts each, using a persistent action ledger and replay safeguards.
- Achieved **100/100 transient-failure recoveries** using bounded retries and persistent checkpoints; **121 automated tests passed** across workflows, persistence, approvals, recovery, and API behavior.

DocWeave | Evidence-Grounded Document Intelligence & Reconciliation GitHub

Stack: *Python, FastAPI, RAG, PostgreSQL, SQLAlchemy, BM25*

- Built 2-way and 3-way procurement reconciliation across purchase orders, invoices, and goods receipts, detecting discrepancies with traceable field-level evidence.
- Benchmarked three retrieval strategies on 13 synthetic queries and 10 evidence chunks; BM25 with query expansion achieved **11/11 Recall@1, 11/11 Recall@3, and MRR 1.00**. All **164/164 tests passed**.

MemoRoute | Temporal-Aware Memory Retrieval & Context Engineering GitHub

Stack: *Python, SQLite, BM25, FastEmbed, BGE, Hybrid Retrieval, RRF*

- Improved Recall@1 from **56.25% to 75%** and reached **93.75% Recall@3** on an 8-query benchmark; hybrid RRF achieved **85.42% MRR** versus 79.17% for BM25.
- Validated temporal reasoning across 20 scenarios and 32 memory records with **168/168 assessment checks complete**; maintained 100% top-3 consistency across six RRF configurations and passed **201 tests**.

TuneBench | GenAI Tool-Calling, Reliability & Evaluation Platform GitHub

Stack: *Python, Qwen3-4B, llama.cpp, JSON Schema, Streamlit, pytest, GitHub Actions*

- Engineered a CPU-run quantized Qwen3-4B workflow with schema-constrained generation, semantic tool-call review, and AST/Decimal refund validation; increased correct tool-call outcomes by **5x** on a matched 40-case synthetic benchmark.
- Built reproducible evaluations with pinned model revisions, hashed datasets, and preserved traces; fixed a review-induced action-escalation defect and validated the application with **561 passing automated tests**.

RESEARCH PUBLICATIONS

Predicting Threatened Habitats Using Logistic Regression ICCCT 2024 · IEEE Xplore

- Built a logistic-regression model using air quality, water pollution, tree-cover loss, CO_2 emissions, urban population, and temperature data sourced from Global Forest Watch, the World Bank, WHO, and Kaggle.
- Achieved **93.55% classification accuracy** using an 80/20 split and 5-fold cross-validation, creating an interpretable early-warning framework for conservation planning. IEEE Xplore

TrustAnchor: Layer-2 Blockchain Provenance for Pharmaceutical Supply Chains SASIGD 2026 · IEEE Xplore

- Designed an offline-first architecture integrating Polygon Layer 2, IPFS, RBAC smart contracts, W3C DID/Verifiable Credentials, and Ed25519 verification.
- Achieved 0.770 ms median offline verification latency, 99.4% on-chain storage reduction, and $0.0008 single-batch anchoring cost; security tests detected QR replay and metadata tampering.

SKILLS

Programming & Languages :	Python, Java, JavaScript, TypeScript, R, SQL, HTML, CSS
Backend :	FastAPI, Flask, Node.js, Express.js, REST APIs, SQLAlchemy, Pydantic, LangGraph, tool calling
Frontend :	React.js
Databases :	PostgreSQL, SQLite, MongoDB
Tools & Platforms :	Git, GitHub Actions, Docker, Ollama, llama.cpp, Streamlit, Alembic, pytest
AI Frameworks & Methods :	NLP, Qwen3-4B, RAG, BM25, BGE, FastEmbed, hybrid retrieval, Reciprocal Rank Fusion (RRF), model evaluation, JSON Schema
Cloud Platforms :	AWS, Oracle Cloud Infrastructure GenAI
Blockchain & Security :	IPFS, Polygon Layer 2, RBAC, prompt-injection defense, W3C DID / Verifiable Credentials, Ed25519

CERTIFICATIONS & LEADERSHIP

AWS Certified Cloud Practitioner - Amazon (issued 27 Jul 2025) · **Oracle Cloud Infrastructure Generative AI Professional** - Oracle Cloud (2025) · **MERN Full Stack** - Ethnus (2024) · **SmartED AI** - Certificate of Internship Completion (AI1332) + Certificate of Course Completion (CCC1721) - Sep 2025 / Jan 2026.

Marketing Team Lead, Climate Change Club - VIT-AP: Led a five-member team and strengthened campus outreach through coordinated sustainability campaigns and promotional content.

EDUCATION

B.Tech, Computer Science and Engineering 2022 - 2026
Vellore Institute of Technology, Andhra Pradesh **Cumulative GPA: 8.02/10.0**

Higher Secondary (10+2), MPC (Maths, Physics, Chemistry) Apr 2020 - Mar 2022
Kendriya Vidyalaya No. 1, Vijayawada CGPA: 8.23

Secondary School (Class X) Apr 2019 - Mar 2020
Kendriya Vidyalaya No. 1, Vijayawada CGPA: 8.95